

prosail: an R package to simulate canopy reflectance with the coupled model PROSAIL (PROSPECT + SAIL)

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Summary

PROSAIL is a vegetation radiative transfer model (RTM) coupling the *leaf optical PRO*PERTIES *SPECTra* (PROSPECT) model, which simulates leaf optical properties ([Jacquemoud & Baret, 1990](#)), with the *Scattering by Arbitrarily Inclined Leaves* model (SAIL) which simulates canopy bidirectional reflectance ([Berger et al., 2018](#); [Jacquemoud et al., 2009](#); [W. Verhoef, 1984](#)) from the visible to shortwave infrared domains. PROSAIL is a valuable tool for agriculture, ecology, and climate research ([Poulter et al., 2023](#)). Inversion procedures allow retrieval of vegetation biophysical properties, including leaf area index (LAI), leaf chemistry, the fraction of absorbed photosynthetically active radiation (fAPAR), and fractional vegetation cover (fCover).

We introduce *prosail*, an R package which provides multiple versions of the model PROSAIL, coupled *prospect* ([Féret & Boissieu, 2024](#)). *prosail* simulates surface reflectance in forward mode for various optical sensors, based on their spectral response function (SRF). It includes functions for model inversion using iterative optimization, and a hybrid approach, combining physical modeling and machine learning (ML) regression.

Statement of need

Vegetation traits mapping and monitoring is crucial to better understand ecosystem functions, carbon, water and energy budgets. Vegetation RTMs describe the interactions between light and vegetation based on their biophysical and chemical properties, through absorption and scattering mechanisms.

The SAIL model is an example of four-stream representations of the radiative transfer equation, including two direct fluxes (incident solar flux and radiance in the viewing direction) and two diffuse fluxes (upward and downward hemispherical flux).

More complex models include the *Soil Canopy Observation of Photosynthesis and Energy fluxes* (SCOPE) ([Yang et al., 2021](#)) model, which features layered description of the vegetation, and the *Discrete Anisotropic Radiative Transfer* DART model ([Gastellu-Etchegorry et al., 2015](#)), which explicitly describes 3D canopy structure.

Hybrid strategies generating a training dataset for ML algorithms with RTM simulations show multiple advantages ([Verrelst et al., 2015](#)): simulations complement or replace in situ sampling for robust estimation, while ML ensures computation efficiency compared to iterative optimization.

The limited number of input parameters and the computational efficiency of SAIL is interesting for operational applications, despite the main assumption of vegetation as a homogeneous

39 turbid medium: leaves are randomly distributed within the canopy, which is not accurate for
40 row crops and heterogeneous canopies.

41 State of the field

42 Various softwares allow hybrid inversion with PROSAIL simulations. The *Sentinel Toolbox*
43 *Application Platform* (SNAP) includes the Biophysical Processor module (Weiss et al.,
44 2020), combining PROSAIL simulations with an artificial neural network regression model.
45 The sampling design of the training set is described in the *Algorithmic Theoretical Background*
46 *Document* (ATBD) (Weiss et al., 2020)). Alternative distributions provide more interactive
47 parameterization of the inversion strategy, e.g. the *Automated Radiative Transfer Models*
48 *Operator* (ARTMO) Matlab toolbox (Caicedo et al., 2014).

49 *prosail* does not intend to provide the same computational efficiency as SNAP. It does
50 not provide a collection of models and methods as comprehensive as those provided with
51 the ARTMO box neither. *prosail* provides a flexible open source framework to experiment
52 with hybrid inversion procedures, using simple yet fast and efficient training stage, allowing
53 experimenting on training data sampling design, introduction of noise, feature selection over
54 optical sensors. It is suitable for research and development, to identify potential and limitations
55 of inversion strategies. Resulting regression models can be applied on remote sensing data,
56 but it may not be the appropriate software for scaling up operational vegetation monitoring
57 applications. Alternative PROSAIL implementations are also available in [python](#), [Julia](#) and [R](#).
58 Other options are available on [this webpage](#).

59 Software design

60 *prosail* uses *prospect* (Féret et al., 2017, 2021) for the simulation of leaf optical properties.
61 It includes *4SAIL* (W. Verhoef et al., 2007), and *4SAIL2* (Wout Verhoef & Bach, 2007), a
62 two layers version of *4SAIL* as canopy models. *prosail* provides user-friendly and modular
63 functions to simulate vegetation canopy reflectance, and to predict biophysical properties using
64 inversion.

65 Hybrid inversion is a multi-step procedure requiring simulation of sensor reflectance to train
66 a ML regression algorithm, then applicable to any data source collected from airborne and
67 spaceborne sensors.

- 68 ■ The generation of realistic reflectance simulations requires definition of input parameter
69 sampling strategy, accounting for sensor SRF and uncertainties. *prosail* provides
70 functions to produce this training dataset as a unique step following standard
71 parameterization, or step by step, in order for users to understand the logic, or adjust
72 each step to their needs.
- 73 ■ A limited number of ML algorithms is currently provided. Additional algorithms will be
74 implemented in future versions.
- 75 ■ Raster processing is handled with the *terra* (Hijmans, 2025) package, for compatibility
76 with most raster data formats.
- 77 ■ *prosail* can be used in combination with [preprocS2](#), a package dedicated to downloading
78 and preprocessing of Earth observation data following the [STAC](#) specification, in order
79 to produce a fully automated image access and processing workflow.

80 Research impact statement

81 *prosail* has been used in multiple research publications since its early developments for
82 applications in agriculture ((Verrelst et al., 2021)), ecology ((Hauser et al., 2021), (Féret et

83 al., 2026)), forestry ((Ferreira et al., 2026)), and vegetation monitoring in general (Kattenborn
84 et al., 2024). The model PROSAIL is used to analyze optical imagery acquired from UAV
85 to airborne to spaceborne sensors, including multispectral satellite (Sentinel-2, Landsat) and
86 imaging spectroscopy sensors (EnMAP, PRISMA, AVIRIS). This package is currently used in
87 multiple research projects, and is actively maintained.

88 Overview

89 Simulating canopy reflectance with PROSAIL

90 PROSAIL requires defining vegetation properties, including leaf optical properties simulated
91 with prospect, or measured experimentally, LAI, and leaf inclination distribution. PROSAIL
92 also requires information extrinsic to vegetation, including the foliage hot spot parameter
93 (Bréon et al., 2002; Kuusk, 1991; Wout Verhoef & Bach, 2007), the sun-observer geometry
94 (sun and observer zenith angles, relative azimuth angle), and soil reflectance. SAIL produces
95 top-of-canopy bi-hemispherical reflectance factor, directional-hemispherical reflectance factor
96 for solar incident flux, hemispherical-directional reflectance factor in viewing direction and
97 bi-directional reflectance factor.

98 These reflectance factors can then be used to simulate surface reflectance acquired by a
99 sensor with known SRF. The SRF from multiple sensors is already implemented in prosail,
100 listed when calling `srf_availability()`. User-defined sensors require either defining central
101 wavelength and full width at half maximum (`fwhm`) for each band, assuming gaussian response,
102 or the exact SRF provided with a `.csv` files.

103 PROSAIL hybrid inversion

104 Simulating surface reflectance to prepare for regression model training

105 Regression models are trained based on i. a sampling strategy (range, distribution, co-
106 distributions for input parameters), ii. a level of noise applicable to the simulated surface
107 reflectance, and iii. spectral bands used for the training of the regression model.

108 The default parameterization of prosail hybrid inversion relies on the production of a surface
109 reflectance LUT compliant with the specifications defined in the ATBD for the Biophysical
110 Processor of the Sentinel toolbox (Weiss et al., 2020). Each step of the simulation of a training
111 LUT can be defined by user.

112 Training a ML regression algorithm

113 The ML strategy implemented in prosail is a parsimonious ensemble method based on a
114 bootstrap aggregating (bagging) prediction of biophysical properties from support vector
115 regression (SVR) models. Each SVR model is trained with a limited number of samples,
116 ensuring fast training stage. The predicted value corresponds to the mean prediction from a set
117 of SVR models. The standard deviation is derived from this ensemble of predictors. Standard
118 deviation may not provide accurate model uncertainty quantification (Palmer et al., 2022).

119 The default parameterization uses an ensemble of 10 SVR models trained with 200 samples
120 each. The training then requires 2000 samples, to be compared to the 41472 simulations used
121 for training models implemented in SNAP. This parsimonious ensemble method ensures fast
122 training stage and retrieval performances similar to SNAP.

123 Performing full hybrid inversion

124 Figure 1 summarizes the workflow applied to perform hybrid inversion. The function
125 `train_prosail_inversion` combines aforementioned steps to simulate surface reflectance

126 and train SVR models. Obtained regression models are then used to estimate vegetation
127 biophysical properties from data tables or raster data.

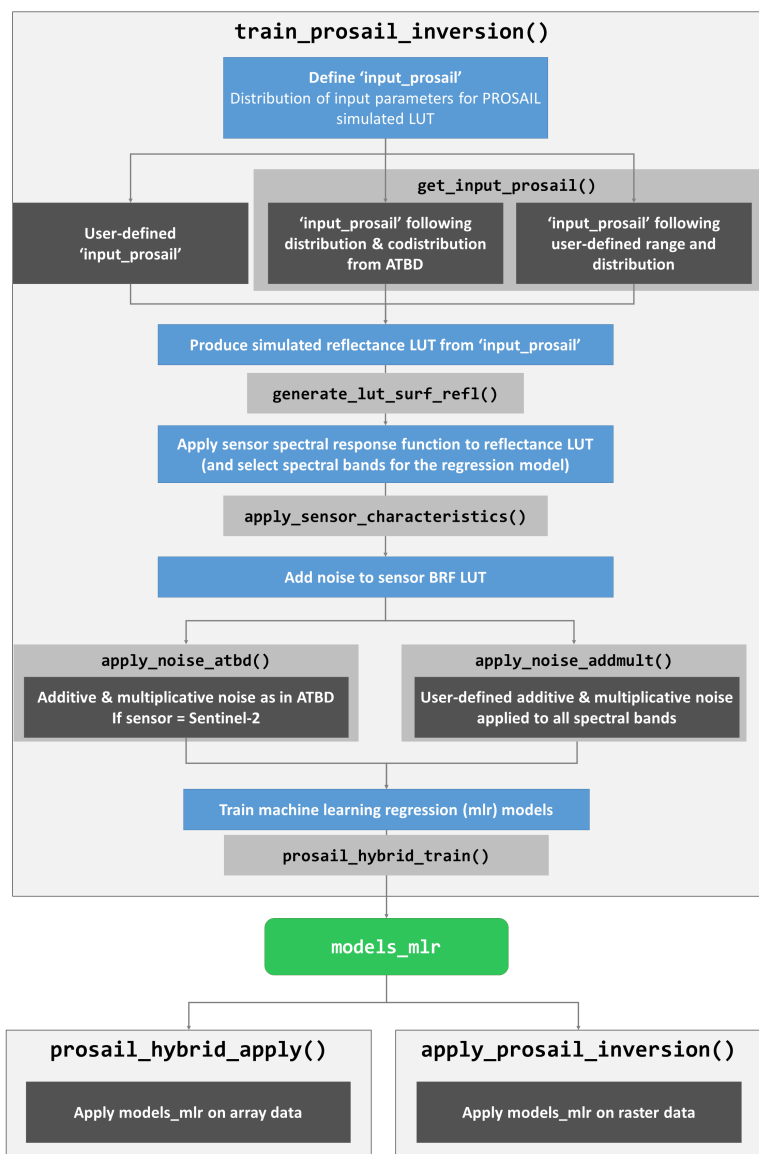


Figure 1: Workflow of PROSAIL hybrid inversion implemented in the package prosail.

128 Application to Sentinel-2 image and comparison with SNAP

129 Once ML regression models are trained, prosail hybrid inversion can be applied on Sentinel-2
130 imagery using the function `apply_prosail_inversion`. Biophysical variables produced with
131 SNAP are compared to those produced with the prosail hybrid inversion. The Sentinel-2
132 Level-2A product corresponding to tile 30SWJ acquired on May 13th, 2021 over an agricultural
133 landscape located next to Barrax (Spain) was downloaded from the Copernicus Data Space
134 Ecosystem (CDSE). LAI, fcover and fAPAR computed from Sentinel-2 images with prosail
135 hybrid inversion are compared to those produced with SNAP in Figure 2.

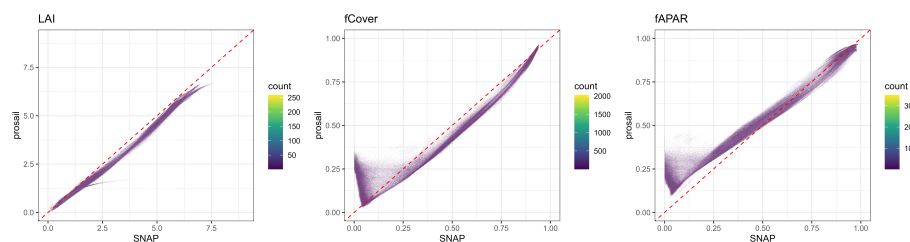


Figure 2: Biophysical properties estimated with *prosail* vs Biophysical properties estimated with SNAP.

The two methods show good consistency, with Pearson correlation coefficient > 0.99 for LAI, and > 0.97 for fCover and fAPAR. Algorithmic differences include the PROSPECT version (SNAP uses a version anterior to PROSPECT-D) and the ML algorithm. Differences in soil properties accounted for in simulations with low LAI may also contribute, despite efforts to reproduce the workflow described in the ATBD. Additional tests performed over croplands and forests showed similar performances. Figure 3 displays a set of biophysical properties mapped over the region of interest.

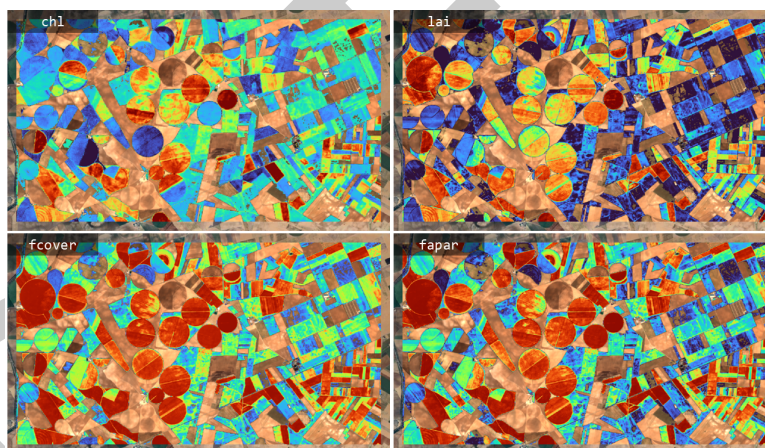


Figure 3: Leaf chlorophyll content, LAI, fCover and fAPAR mapped over the region of Barrax (Spain).

Conclusion

prosail is an R package dedicated to the canopy reflectance model PROSAIL, simulating reflectance from optical sensors based on their spectral response. It includes inversion based on iterative optimization and ML/RTM inversion. Vegetation biophysical properties estimated with a hybrid inversion are consistent with estimations from SNAP. *prosail* hybrid inversion does not intend to be computationally as efficient as SNAP. The current version not appropriate for regional to global scale vegetation monitoring. *prosail* hybrid inversion offers a fully adjustable hybrid inversion framework, including an original parsimonious ML regression method, allowing fast and efficient training. It is a valuable tool for experimenting on the potential and limitation of physically-based retrieval of vegetation traits from optical sensors, including satellite missions already operational and in preparation.

Availability

prosail is an open-source software made available under the GNU GPL-3 license. Tutorials are available at <https://jbferet.gitlab.io/prosail/>.

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AI Usage Disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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